

QUALIFYING EXAMINATION, Part 3

Solutions

Problem 1: Quantum Mechanics I

(a) There is degeneracy of $2l + 1$ in the magnetic quantum number and 2 in spin for a total degeneracy of $2(2l + 1)$.

The degeneracy is due to the rotational symmetry of the system, with respect to orbital and spin angular momentum independently.

The values of j are $j = l \pm \frac{1}{2}$ for $l \neq 0$ and $j = \frac{1}{2}$ for $l = 0$.

The degeneracy of each j is $2j + 1$. The total degeneracy is then

$$2(l + \frac{1}{2}) + 1 + 2(l - \frac{1}{2}) + 1 = 4l + 2 = 2(2l + 1).$$

(b) Using $\vec{L} \cdot \vec{S} = \frac{1}{2}(\vec{J}^2 - \vec{L}^2 - \vec{S}^2)$, we have

$$\langle \vec{L} \cdot \vec{S} \rangle = \frac{\hbar^2}{2}[j(j + 1) - l(l + 1) - 3/4].$$

Thus

$$\begin{aligned} \langle \vec{L} \cdot \vec{S} \rangle &= (\hbar^2/2)l & j = l + 1/2, \\ \langle \vec{L} \cdot \vec{S} \rangle &= -(\hbar^2/2)(l + 1) & j = l - 1/2. \end{aligned}$$

(c) Using the results in part (b), we have

$$\begin{aligned} \langle H_{LS} \rangle &= \frac{\hbar^2}{4mc^2}l \langle \frac{1}{r} \frac{dV}{dr} \rangle & j = l + \frac{1}{2}, \\ \langle H_{LS} \rangle &= -\frac{\hbar^2}{4mc^2}(l + 1) \langle \frac{1}{r} \frac{dV}{dr} \rangle & j = l - \frac{1}{2}. \end{aligned}$$

There are two levels with degeneracy $2l + 2$ for $j = l + 1/2$ and $2l$ for $j = l - 1/2$.

The degeneracy remains due to the rotational symmetry of the system with respect to total angular momentum.

(d) The d state is split by the fine structure into two levels: 6 $j = 5/2$ states and 4 $j = 3/2$ states. The $j = 5/2$ states lie above the $j = 3/2$ states.

Problem 2: Quantum Mechanics II

(a) The Hamiltonian is invariant under rotations around the z axis, so $[\hat{H}, \hat{L}_z] = 0$. Rotations around x and y are broken by the ionic potential, so we expect $[\hat{H}, \hat{L}_x] = [\hat{H}, \hat{L}_y] \neq 0$. The two commutators are the same, because $\hat{L}_x$ and $\hat{L}_y$ are related by a $\pi/2$ rotation around the z axis.

(b) $\hat{L}_z$ generates the $\text{SO}(2) \simeq \text{U}(1)$ group. The group is Abelian and has only 1-dimensional irreducible representations, so by itself does not imply any degeneracy in the spectrum.

(c) Since $[\hat{H}, \hat{L}_z] = 0$, the eigenstate of $\hat{H}$ can be chosen to be an eigenstate of $\hat{L}_z$ as well. In the cylindrical coordinate, $\hat{L}_z = -i\frac{\partial}{\partial\varphi}$, whose eigenfunctions are $e^{im\varphi}$ with $m \in \mathbb{Z}$. When $\hat{L}_z$ is fixed to be m , the Hamiltonian in cylindrical coordinate becomes

$$\hat{H}_m = -\frac{1}{2m_e} \left[\frac{1}{\rho} \frac{\partial}{\partial\rho} \left(\rho \frac{\partial}{\partial\rho} \right) - \frac{m^2}{\rho^2} + \frac{\partial^2}{\partial z^2} \right] - \frac{e^2}{\sqrt{\rho^2 + (z-d)^2}} - \frac{e^2}{\sqrt{\rho^2 + (z+d)^2}},$$

whose eigenfunctions are then denoted by $f_{\alpha m}(\rho, z)$. Thus we found the eigenfunctions of $\hat{H}$ to be of the form $\psi_{\alpha m}(\rho, z, \varphi) = f_{\alpha m}(\rho, z)e^{im\varphi}$.

(d) In the Cartesian coordinate, $\nabla^2 = -(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2})$, which is obviously invariant under $x \rightarrow -x$. The potential energy is a function of ρ and z , both of which are invariant $\hat{P}_x$. Therefore $\hat{H}$ is invariant under $\hat{P}_x$, i.e., $[\hat{H}, \hat{P}_x] = 0$.

(e) Under $\hat{P}_x$ we have

$$\hat{P}_x : \hat{x} \rightarrow -\hat{x}, \quad \hat{p}_x \rightarrow -\hat{p}_x, \quad \hat{y} \rightarrow \hat{y}, \quad \hat{p}_y \rightarrow \hat{p}_y.$$

Therefore $\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \rightarrow -\hat{L}_z$ under $\hat{P}_x$. This means $\hat{P}_x\hat{L}_z\hat{P}_x^{-1} = -\hat{L}_z$ or $\hat{P}_x\hat{L}_z = -\hat{L}_z\hat{P}_x$.

(f) Since $\hat{L}_z$ changes sign under $\hat{P}_x$, $\hat{P}_x\psi_{\alpha m}$ becomes $\psi_{\alpha, -m}(\rho, z, \varphi) = f_{\alpha m}e^{-im\varphi}$. Because $[\hat{H}, \hat{P}_x] = 0$, $\psi_{\alpha, -m}$ is also an energy eigenstate and has the same energy as $\psi_{\alpha m}$.

(g) To first order in degenerate perturbation theory, the effective Hamiltonian is just the projection of the perturbation to the degenerate subspace. Here we are concerned with

the subspace spanned by $\psi_{\alpha,+1}$ and $\psi_{\alpha,-1}$. Let us compute the general matrix element:

$$\begin{aligned}
\langle \psi_{\alpha,n} | \hat{x} | \psi_{\alpha',n'} \rangle &= -e\mathcal{E} \int \rho d\rho dz \rho f_{\alpha n}^*(\rho, z) f_{\alpha' n'}^*(\rho, z) \int_0^{2\pi} d\varphi e^{i(n'-n)\varphi} \cos \varphi \\
&= -e\mathcal{E} \int \rho d\rho dz \rho f_{\alpha n}^*(\rho, z) f_{\alpha' n'}^*(\rho, z) \frac{1}{2} \int_0^{2\pi} d\varphi e^{i(n'-n)\varphi} (e^{i\varphi} + e^{-i\varphi}) \\
&= -\pi e\mathcal{E} (\delta_{n'-n,1} + \delta_{n'-n,-1}) \int \rho d\rho dz \rho f_{\alpha n}^*(\rho, z) f_{\alpha' n'}^*(\rho, z).
\end{aligned}$$

It follows that

$$\langle \psi_{\alpha,\pm 1} | \hat{x} | \psi_{\alpha,\pm 1} \rangle = \langle \psi_{\alpha,\pm 1} | \hat{x} | \psi_{\alpha,\mp 1} \rangle = 0.$$

Thus the energy splitting vanishes to order $O(\mathcal{E})$